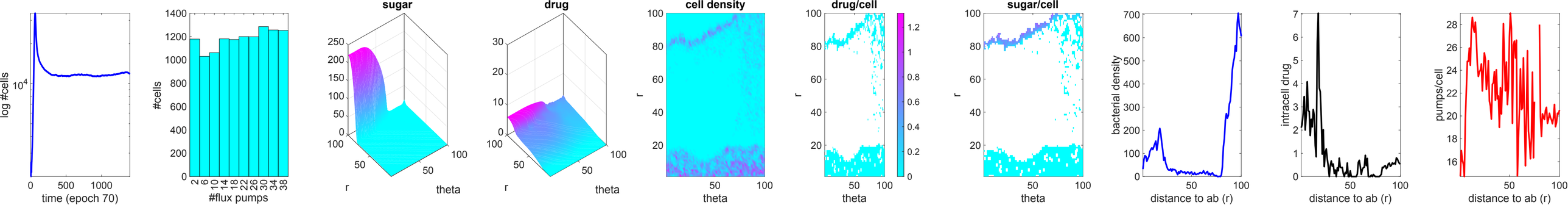
log #cells
10^4
0
500
1000
time (epoch 70)
#cells
1400
1200
1000
800
600
400
200
0
2 6 10 14 18 22 26 30 34 38
#flux pumps
sugar
250
200
150
100
50
0
100
50
r
50
100
theta
drug
30
20
10
0
100
50
r
50
100
theta
cell density
100
80
60
40
20
r
50
100
theta
drug/cell
100
80
60
40
20
r
50
100
theta
1.2
1
0.8
0.6
0.4
0.2
0
sugar/cell
100
80
60
40
20
r
50
100
theta
bacterial density
700
600
500
400
300
200
100
0
50
100
distance to ab (r)
intracell drug
7
6
5
4
3
2
1
0
50
100
distance to ab (r)
pumps/cell
28
26
24
22
20
18
16
50
100
distance to ab (r)